



Q A 55-year old male presents to the clinic with a history of progressively worsening headaches over the past few months and multiple episodes of focal seizures. He has multiple lumps all over the body. MRI shows single ring enhancing lesion in the left parietal cortex with contrast enhancement. Which is next best investigation to be done?

- a. CSF EITB for antibodies to cysticerci
- b. CSF NAT for MTB
- c. Magnetic resonance spectroscopy
- d. Diffusion weighted MRI

Ans: (a)

Q. Match the given syndromes with the expected intracranial lesions:

1. Tuberous sclerosis	a. Cerebellar hemangioblastoma
2. Von Hippel Lindau	b. Subependymal astrocytoma
3. Cowden Syndrome	c. Cerebellar ganglioblastoma

- a. 1-a; 2-c; 3-b
- b. 1-a; 2-b; 3-c
- c. 1-c; 2-a; 3-b
- d. 1-b; 2-a; 3-c

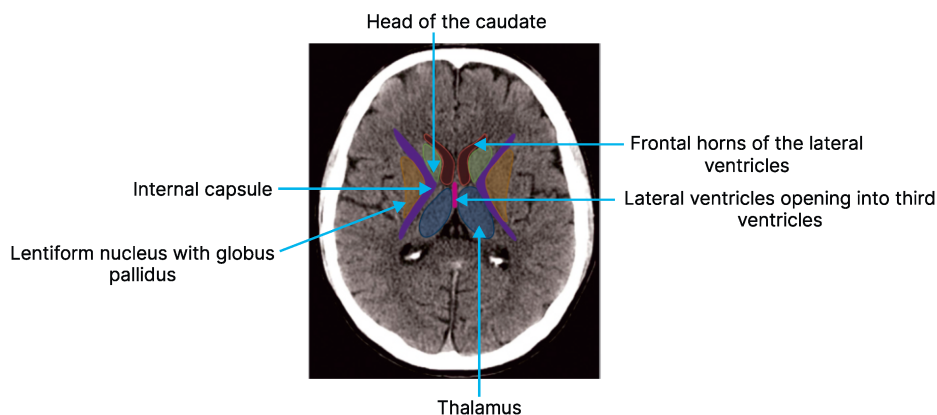
Ans: (d)



INTRACRANIAL HEMORRHAGE

INTRACEREBRAL HEMORRHAGE/ INTRAPARENCHYMAL HEMORRHAGE BASIC NEUROANATOMY

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- Involvement of Caudate nucleus - **chorea**
- Internal capsule - consists of anterior and posterior limb, corticospinal pathway transverses through it
- Lentiform nucleus consists of
 - Globus pallidus
 - Putamen - **M/c** site of intracerebral hemorrhage

CAUSES

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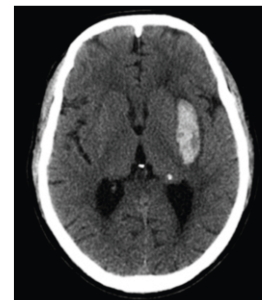
- 10% of all strokes
- **Hypertension** (M/c)- hypertension crisis
 - Damage to penetrating branches of lenticulate arteries
- Drugs - Warfarin/Apixaban/Dabigatran toxicity
- Cocaine (potent vasoconstriction), Methamphetamine
- Arteriovenous malformation
- Elderly population - Lobar hemorrhage (No h/o smoking, DM II, HTN)
 - Cerebral amyloid angiopathy (Fragile Blood vessel)
 - A β amyloid deposition (Apolipoprotein E gene)
- Trauma
- Moya Moya disease - Occlusive arterial disease
 - Young population - Intracranial hemorrhage risk
 - **Angiography** - Puff of smoke like appearance

CASE SCENARIO

Q. 50-year-old man with hypertension working as a senior manager in a bank is taking a sales meeting. While speaking, he develops sudden onset of face deviation, slurring of speech and weakness in his right arm and right leg over next 1 hour. On examination his both eyes shows conjugate gaze deviation to left side. NCCT head is shown below.



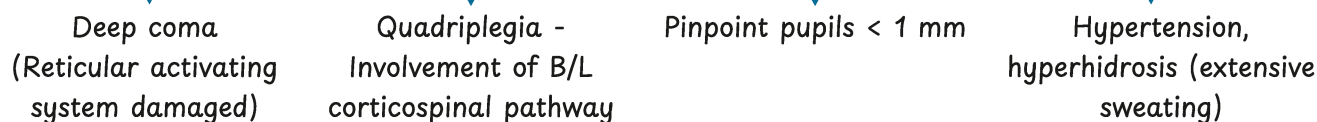
- Slurring of speech
- Weakness in right arm and leg
- Conjugate gaze to left side- involvement of Left middle cerebral artery and its branches
 - Cortical stroke - gaze towards side of lesion
 - Brainstem stroke - gaze away from side of lesion
- NCCT - hyperdensity on left side



DIAGNOSIS

- Intracerebral hemorrhage of left side
- M/c site of Hemorrhagic stroke/ Intracranial hemorrhage- Putamen

PONTINE HEMORRHAGE



Important Information

- Pinpoint pupils also seen in heroin overdose - absent hyperhidrosis, respiratory depression present

ICH SCORE (0-6)

- GCS - 5 - 12 - score of 1, 3 - 4 - score of 2
- Infratentorial origin of hemorrhage present
- Intraventricular hemorrhage present (extension of intracerebral hemorrhage)
- Hematoma volume > 30 cc
- Age > 80 years
- Higher score → bad prognosis

Important Information

- Intraventricular hemorrhage -
 - Neonates → birth injury (faulty forceps)
 - Shrill cry, Bulging anterior fontanelle
 - IOC - Skull USG

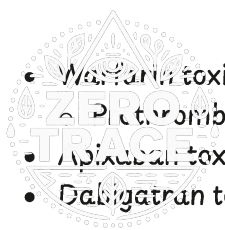
MANAGEMENT OF INTRACEREBRAL HEMORRHAGE

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- No role of Alteplase/ Tenecteplase
- Hypertensive crisis > 220/ 120 mmHg + End organ damage
 - Target BP < 140 mmHg
 - Drug used - Nicardipine (DOC)/ Labetalol

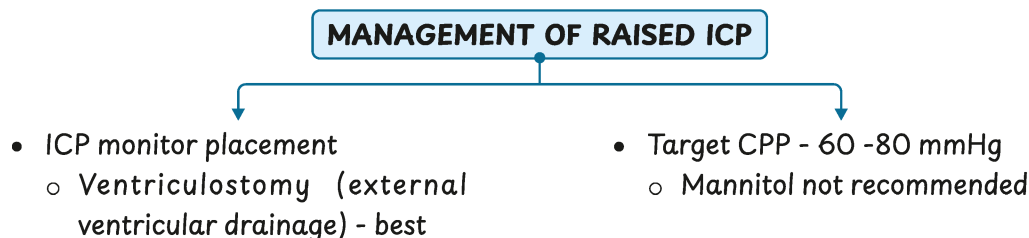
Important Information

- DOC in Cardiovascular hypertensive crisis → Nitroglycerine



- Warfarin toxicity
 - Prothrombin complex concentrate + Vitamin K > FFP
- Apixiban toxicity - **Andexanet alpha**
- Dabigatran toxicity- Idarucizumab

MANAGEMENT OF RAISED ICP



CASE SCENARIO

A boy on his 25th birthday was traveling on his new bike without a helmet. Bike skidded on wet road and his head slammed against the concrete road. He became unconscious for few minutes and then regained consciousness and noticed he had severe left knee pain and multiple abrasions. He came to hospital for treatment by himself. The doctor told nurse to administer first aid and made him rest on a bed. After 10-15 mins, when Doctor examined the patient, GCS is 8/15. Urgent NCCT head shows lenticular hyperdensity.

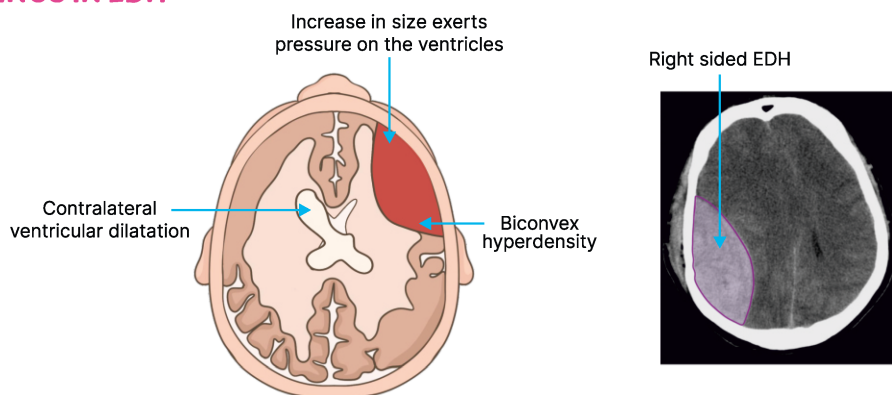
INFERENCE

- State of unconsciousness with consciousness in between lucid interval
 - Lucid interval seen in - Extradural/epidural and Subdural hemorrhage
- CT- lenticular hyperdensity (biconvex hyperdensity)

DIAGNOSIS

- Extradural hemorrhage (Epidural hemorrhage)

NCCT HEAD FINDINGS IN EDH



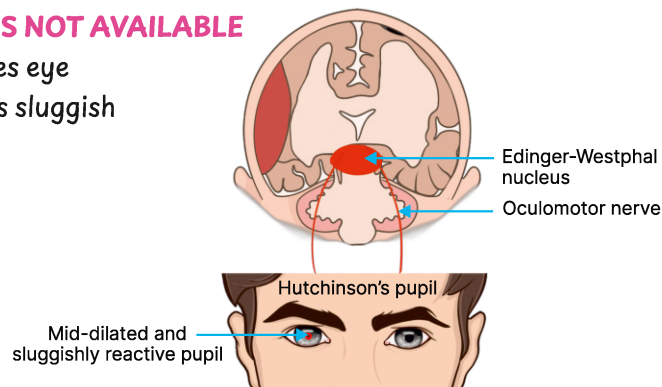
- Source of bleeding in EDH - **Middle meningeal artery** (branch of ECA)

T/t

- Rapid surgical evacuation by **craniotomy**, Ligation of bleeder
- Burr hole surgery for EDH with posturing
 - Site - Pterion

DECISION OF SIDE OF BURR HOLE IN EDH IF CT SCAN IS NOT AVAILABLE

- Edinger-Westphal nucleus via Oculomotor nerve supplies eye
- Expanding hematoma ↑ pressure on affected side shows sluggish pupillary reflex to light
- Sluggish pupillary reflex to light → **Hutchinson's pupil**
- Burr hole done on same side of Hutchinson's pupil



CASE SCENARIO

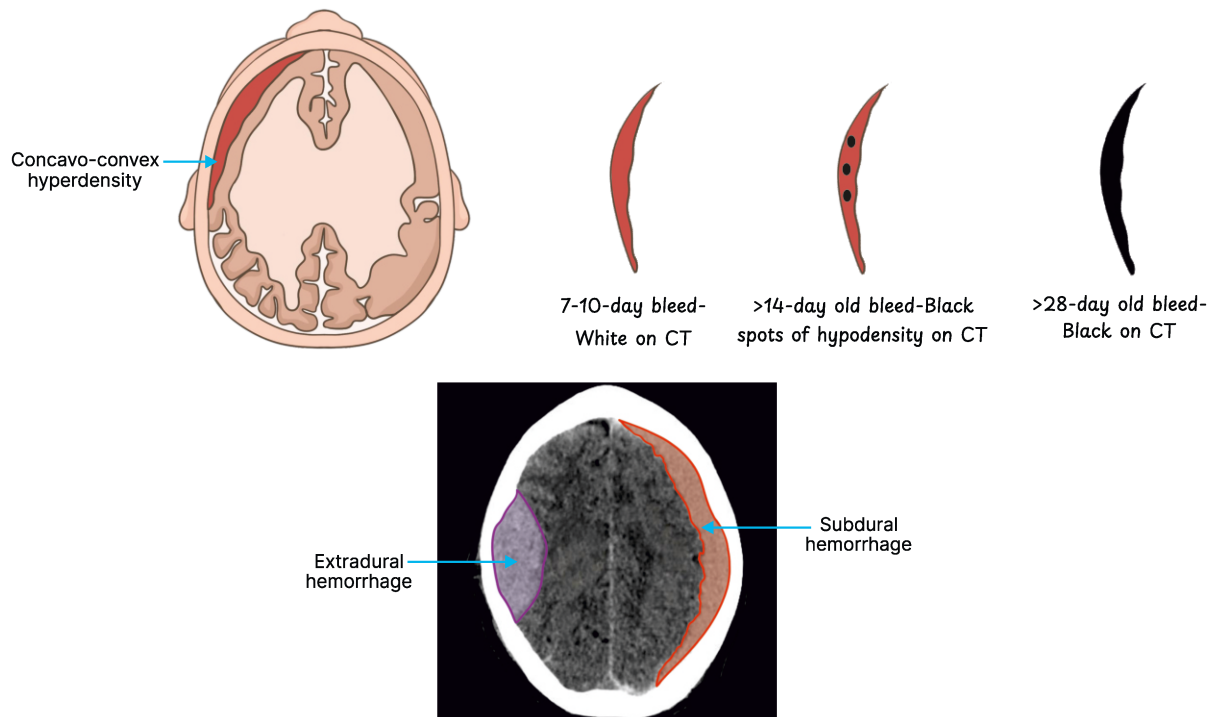
70 year old obese female with Diabetic neuropathy slipped in the bathroom and her head slammed against the bathroom floor. She noticed immediate bump on her forehead but did not visit the hospital on day of fall. On the next day her son notices that his mother is becoming irritable, inattentive and feels drowsy all the time along with incoherence. 2 weeks after sustaining the fall she did not wake up at all, her son rushes her to hospital. GCS on admission 8/15 with brisk DTR and extensor plantars. NCCT head shows crescentic hypodensity pressing on right frontal and temporal bone.

INFERENCE

- Gradual development of irritability, drowsiness and incoherence
- NCCT - crescentic hypodensity

DIAGNOSIS

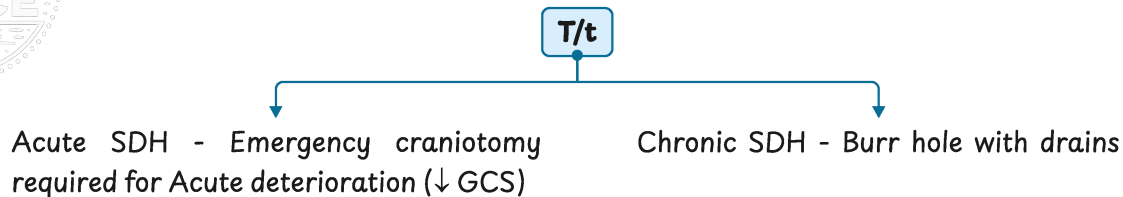
- Chronic subdural hemorrhage (SDH)



- Lucid interval seen in Acute SDH, EDH
- Source of bleeding in SDH - **Cortical bridging veins**



Star SDH may be asymptomatic and do not require surgical evacuation

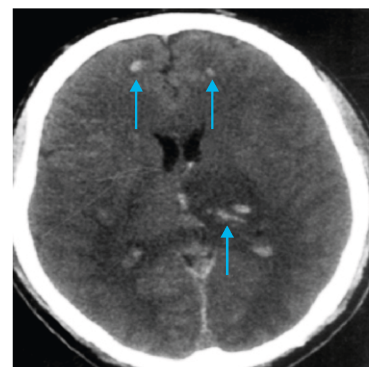


Important Information

- Burr hole can be done in any patient with CNS bleed

CASE SCENARIO

Q. A 25-year-old guy driving a car jumps a red light and has an accident. He is admitted to your hospital with polytrauma for past 3 days. GCS today morning is 8/15. NCCT head shown below. Diagnosis is?



INFERENCE

- NCCT - **Punctuate hyper densities** present at Gray-White matter interface

DIAGNOSIS

- Traumatic axonal injury
 - For diagnosis of Diffuse axonal injury (DAI) ≥ 4 **microbleeds**

Important Information

- In some cases of DAI, NCCT head may show normal findings
- IOC** - DAI/ TAI $\rightarrow$ MRI head

TYPE OF CNS BLEED	SOURCE OF BLEEDING	CT HEAD FINDINGS
INTRA CEREBRAL	Penetrating branches of lenticulostriate artery	Ill-defined bleed (Putamen)
EXTRADURAL/ EPIDURAL	Middle meningeal artery	Biconvex/Lenticular hyperdensity
SUBDURAL	Cortical bridging veins	Crescentic lesion - Hyperdensity lesion - Acute SDH - Hypodensity lesion - Chronic SDH
SUBARACHNOID	Middle cerebral artery	Star of death lesion/Blood in Sylvian fissure



Q. Which of the following is the leading cause of lobar brain hemorrhage in elderly?

- a. Cerebral amyloid angiopathy
- b. Berry aneurysm
- c. Head trauma
- d. Cavernous angioma

Ans: (a)



32. SYSTEMIC LUPUS ERYTHEMATOSUS

INTRODUCTION

- Skin manifestations: Malar rash (**Malar flush** is seen in **Mitral stenosis**)
 - Involves nose, cheeks, and ear lobes
 - Associated with photosensitivity (Leads to first-degree skin burns)
- **Type 3 hypersensitivity** reaction- Causes ↑ Autophagy, Netosis (Neutrophil extracellular traps which self-damage blood vessels, especially in the kidney, causes Lupus nephritis)
- Females > Males (5.5-6.5:1)
- Gene involved: **TREX-1** (X chromosome)
- Severity of SLE in Klinefelter syndrome (47 XXY): ↑
- M/c body system involved in SLE: **Musculoskeletal system**
- Maximum value assigned to which body system involved in the 2019 EULAR /ACR criteria: **Kidney** (Lupus nephritis)
- **≥ 10 points** must be accumulated for diagnosis of SLE as per EULAR/ACR criteria
 - Spread over **7** clinical domains and **3** immunological domains
- Obligatory entry-level criteria to say it is SLE: **+ve ANCA** (>1:80 titre on Hep-2 cells)



Malar rash

SLICC CRITERIA

00:09:34

MNEMONIC: **SONAR**

- For diagnosis of SLE
 - At least 4 of the following clinical and immunological features need to be present
 - Out of which at least 1 clinical and 1 immunological feature should be present (1 clinical + 3 immunological or 3 clinical +1 immunological)

CLINICAL CRITERIA

SYNOVITIS

- Joint involvement: **≥2**
- Symptoms:
 - Soft tissue swelling
 - Tenderness over the small joints
 - Knee joint involvement

SEROSITIS

	LUNGS	HEART
M/c manifestations	• Pleuritis ◦ Pleural rub	• Pericarditis (ECG: ST elevation) ◦ Pericardial friction rub



Manifestations

- Shrinking lung syndrome
- **Libman sacks endocarditis** (affects Mitral valve)
- Myocarditis
- Arrhythmias
- Accelerated atherosclerosis (↑ incidence of CV mortality)

- To differentiate Pericardial friction from Pleural rub
 - Breath holding → Sound persists despite breath holding → Pericarditis

SKIN INVOLVEMENT

- Subacute cutaneous lupus erythematosus (SCLE)
 - Photosensitivity (**Anti Ro Ab**), Malar rash (Butterfly rash)
- Chronic cutaneous lupus erythematosus
 - Discoid lupus
 - Disc like lesions with hyperpigmentation in periphery and hypopigmentation in centre
 - **Atrophic scarring** (D/t damage of sweat and sebaceous glands)
 - Panniculitis (Subcutaneous tenderness)



Subacute cutaneous lupus erythematosus



Discoid Lupus Erythematosus

ORAL/NASAL ULCERS

- Aphthous ulcer (Also seen in stress, IBD, Bechet's disease)
- In SLE, **no genital involvement**

Important Information

- Bechet's disease have both oral and genital ulcer
 - In females- vulva
 - In males- Scrotum

NONSCARRING ALOPECIA

NEUROLOGICAL MANIFESTATIONS

- Headache
 - Seizures/ epilepsy (Lupus cerebritis)
 - Myelopathy (Rare)
 - Cranial nerve/Peripheral nerve involvement
 - Depression
 - Psychosis
- } **Anti-ribosomal P antibody**

AUTOIMMUNE HEMOLYTIC ANEMIA

• Coombs test: +ve

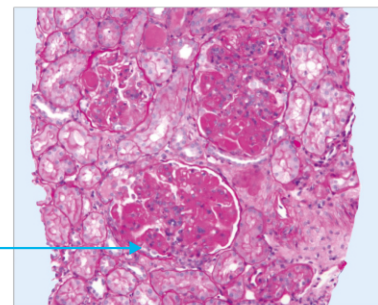
- Leukopenia, Lymphopenia, Thrombocytopenia (Petichiae)
- On peripheral smear: Spherocytes
- T/t: Steroids

Important Information

- M/c type of anaemia in SLE: **Anaemia of chronic disease** (Normocytic normochromic anemia)

RENAL INVOLVEMENT

- Urine spot sample: Urinary protein to creatinine ratio (UPCR)- **>0.5** (Normal: <0.2)
- Presence of RBC casts: Haematuria
- Biopsy findings:
 - **Wire loop lesions** (On light microscopy)
 - Class III & IV nephritis
 - Immunofluorescence finding: Full house effect
- Decreased GFR values:
 - Increased renin values
 - Hypertension
 - Renal failure (eGFR <15 ml/min/1.73m² BSA)



Thickened capillary loops are visible

Renal Cortex biopsy: Lupus Nephritis

IMMUNOLOGICAL FEATURES

Anti-nuclear antibody	Most sensitive test
Anti-dsDNA	Monitors the severity of the disease
Anti-Smith antibody	Most specific test
Anti-phospholipid antibody	Lupus anticoagulant → Placental infarction (causes recurrent abortions, Intrauterine death), False positive RPR (VDRL)
Low serum complements	↓C ₃ , ↓C ₄ , ↓Ch ₅₀
Coombs test	+ ve

CAUSES OF MORTALITY

00:35:59

- Leading cause of death after 45 yrs of age: **Accelerated atherosclerosis**
- Leading cause of death in 1st decade of diagnosis: **Nephritis and infection**
 - For Class 1 & 2 nephritis → Monitors for progression of disease
 - For Class III: Focal lupus nephritis
 - For class IV: Diffuse lupus nephritis
 - For class V: Membranous lupus nephritis
 - For class VI: Dialysis and kidney transplantation

Tacrolimus & Mycophenolate

Entry Level criterion

Antinuclear antibodies (ANA) at a titer of $\geq 1:80$ on Hep-2 cells or an equivalent positive test (ever)



If absent, do not classify as SLE

If present, apply additive criteria



Additive criteria

Do not count a criterion if there is a more illoely explanation than SLE

Occurrence of a criterion on at least one occasion is sufficient

SLE classification requires at least one clinical criterion and ≥ 10 points

Criteria need not occur simultaneously

Within each domain, only the highest weighted criterion is counted toward the total score

Clinical domains and criteria	Weight	Immunology domain and criteria	Weight
Constitutional Fever	2	Antiphospholipid antibodies Anti-cardiolipin antibodies OR Anti-b2GP1 antibodies OR Lupus anticoagulant	2
Hematologic Leukopenia Thrombocytopenia Autoimmune hemolysis	3 4 4	Complement proteins Low C3 OR low C4 Low C3 AND C4	3 4
Neuropsychiatric Delirium Psychosis Seizure	2 3 5	SLE - specific antibodies Anti-ds DNA antibody OR Anti-Smith antibody	6
Mucocutaneous Non-scarring alopecia Oral ulcers Subacute cutaneous OR discoid lupus Acute cutaneous lupus	2 2 4 6		
Serosal Pleural or pericardial effusion Acute pericarditis	5 6		
Musculoskeletal Joint Involvement	6		
Renal Proteinuria >0.5 g/24 h Renal biopsy Class II or V lupus nephritis Renal biopsy Class III or IV lupus nephritis	4 8 10		



Total score:



Classify as Systemic Lupus Erythematosus with a score of 10 or more if entry criterion fulfilled.

AUTOANTIBODIES IN SLE

00:41:32

Anti-nuclear antibody	• >1:80 on Hep-2 cells (Entry level criteria) • Most sensitive test • Best screening test
Anti-ds DNA antibody	• High titres are specific, correlate with disease activity
Anti-smith antibody	• Most specific test
Anti C1q	• Seen >60% of cases • Indicates the severity of Lupus nephritis
Anti-Ro antibody	• Causes photosensitivity • Cross placenta → Neonatal lupus (Affects heart and damage AV node → Complete heart block)
Anti-La antibody	• Dilute Russel viper venom test (Patient's Blood sample + Russel viper venom) → Clot formation indicates placental infarction
Anti-histone antibody	• Associated with drug-induced lupus (Brain and kidneys are spared) ◦ Caused by SHIP drugs (Sulfonamides, Hydralazine, Isoniazid, Procainamide (M/c)) → Procainamide is used in the management of Wolff Parkinson white syndrome ◦ Good prognosis
Anti- phospholipid antibody	• ELISA Cardiolipin test • $\beta 2$ Glycoprotein ELISA test
Anti-erythrocyte antibody	• Cause Autoimmune haemolytic anaemia
Anti-platelet antibody	• Causes epistaxis, menorrhagia, petechiae
Anti-neuronal antibody	• Causes Lupus cerebritis
Anti-ribosomal-P antibody	• Causes neuropsychiatric manifestations

PREGNANCY AND LUPUS

00:49:33

- T/t: For Active SLE in pregnancy
 - **Hydroxychloroquine** + Prednisolone
 - Steroids are not used as a 1st line therapy due to
 - Placental enzyme: 11 β dehydrogenase → **Deactivates steroids**
 - Fetal risk



TREATMENT FOR ANTIPHOSPHOLIPID ANTIBODY SYNDROME

- **Low dose** Corticosteroid therapy → Placental infarction
- **Low dose Aspirin** + Low molecular weight Heparin (**Enoxaparin**)

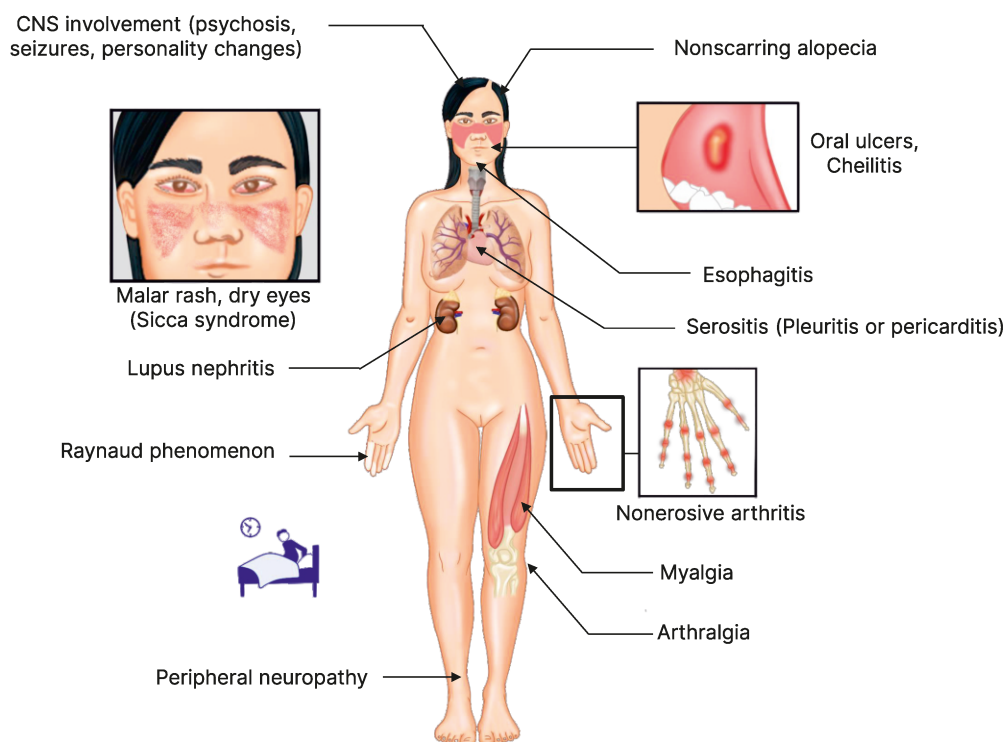
TREATMENT

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TREATMENT

- Not life/organ threatening
 - **Low dose Glucocorticoids**
 - Consider **Belimumab** in Steroid toxicity
- Life/organ threatening
 - **High dose Glucocorticoids** + Mycophenolate/Cyclophosphamide
 - Lupus nephritis (For class III, IV & V)
 - Mycophenolate + Tacrolimus

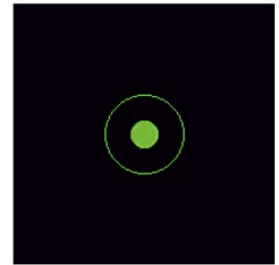
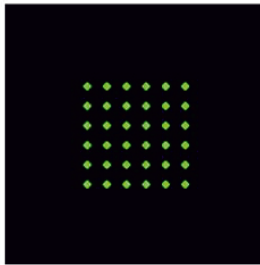
MANIFESTATIONS OF SLE



ANA PATTERNS

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PATTERN	ANTIBODY	ASSOCIATED DISEASE	MNEMONIC
Homogenous	Anti-dsDNA, Anti-histone	SLE, Drug-induced lupus	Diffuse histone haze
Speckled	Anti-Sm, Anti-RNP, Anti-SSA/SSB	SLE, Sjogren's, MCTD	Specks for Sm + SSA
Nucleolar	Anti-fibrillarin, Anti-PM/Scl	Systemic sclerosis (Diffuse)	Nucleoli= Scleroli
Centromere	Anti-centromere	CREST syndrome	C= CREST, Centromere
Rim (Peripheral)	Anti-dsDNA	SLE (Esp. active disease)	Rim= DNA ring



Homogeneous

Disease: SLE, Drug-induced lupus

Pattern: Smooth uniform staining of entire nucleus

Speckled

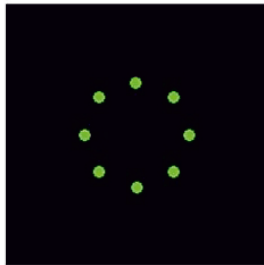
Disease: SLE, Sjögren's, MCTD

Pattern: Fine or coarse speckles in nucleus

Nucleolar

Disease: Systemic sclerosis

Pattern: Bright staining of nucleoli



Centromere

Disease: CREST syndrome

Pattern: Discrete speckles, like dots in mitotic cells



Rim (Peripheral)

Disease: SLE (anti-dsDNA)

Pattern: Bright ring at the nuclear periphery

ANTIPHOSPHOLIPID ANTIBODY SYNDROME

00:58:39

- Antibody-mediated thrombophilia
- Can be primary or secondary (With SLE)
- Catastrophic APS: Involvement of ≥ 3 organs
- When to suspect
 - <55 yrs old with CVA
 - Thrombosis (DVT) → Budd Chiari syndrome, Pulmonary thromboembolism
 - Repeated pregnancy losses
- Antibodies cause increased apoptosis of endothelial cells: Activation of clotting cycle
- Pregnancy morbidity
 - Unexplained death of a morphologically normal fetus >10 weeks
 - Preterm baby: <34 weeks, Pre-eclampsia, PIH
 - >3 unexplained consecutive spontaneous abortions

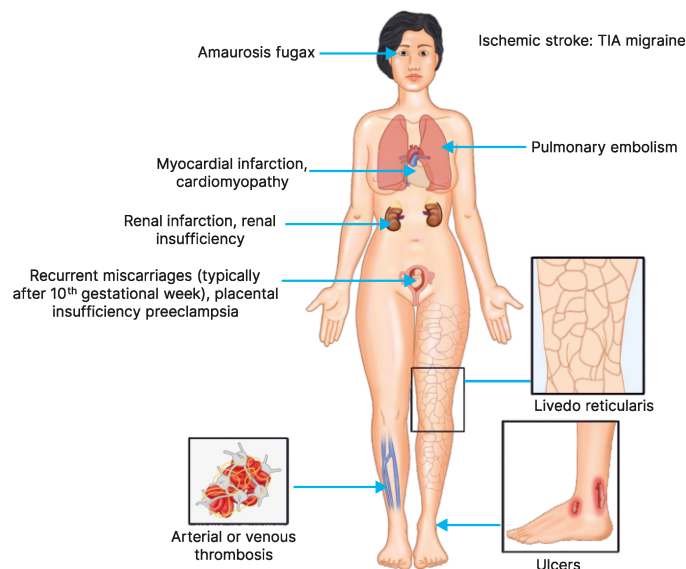
WORKUP

- Lupus anticoagulant test (*Dilute Russell viper venom test*)
 - aPTT: ↑
 - Kaolin clotting time: ↑
- ELISA Cardiolipin antibody tests
- $\beta 2$ Glycoprotein ELISA test

ANTIPHOSPHOLIPID SYNDROME: DIAGNOSTIC CRITERIA COMPARISON

Feature	2006 Sapporo (Sydney Criteria)	2023 ACR/EULAR Criteria
Purpose	Classification & diagnosis (widely used in clinical settings)	Primarily classification (for research)
Entry Criterion	None specified	≥ 1 positive aPL test within 3 years
Clinical Criteria	1. Vascular thrombosis 2. Pregnancy morbidity	6 Clinical Domains: • Macrovascular thrombosis • Microvascular thrombosis • Obstetric morbidity • Cardiac valve disease • Hematologic features • Neurological events
Laboratory Criteria	1. Lupus anticoagulant 2. Anti-cardiolipin (IgG/IgM) 3. Anti-β2GPI (IgG/IgM) (2 positive results 12 weeks apart)	Focused on: • Lupus anticoagulant (functional tests) • Solid-phase ELISA for aCL & anti-β2GPI
Timing of Lab Tests	Tests repeated at least 12 weeks apart	aPL positivity any time within 3 years of the clinical event
Use in Practice	Standard for diagnosis in clinical medicine	For classification; clinical judgment still essential for diagnosis
Publication Reference	Miyakis et al., 2006	Cervera et al., ACR/EULAR 2023

MANIFESTATIONS OF ANTIPHOSPHOLIPID ANTIBODY SYNDROME



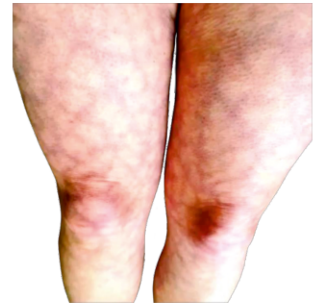


LIVID Reticularis

• Appearance

• Causes

- SLE
- Antiphospholipid antibody syndrome
- Polyarteritis nodosa
- Cryoglobulinemia



MCQ

01:05:36

Q. A 19-year-old female recently diagnosed with SLE has been admitted to the hospital. She is being tested for various autoantibodies related to SLE. Which of the following statements about the specific autoantibodies in SLE is incorrect?

- Anti-smith antibody is highly specific
- Anti-Ro (SS-A) antibody is associated with photosensitivity and can cause complete heart block in fetus
- Anti-dsDNA antibody correlates with disease activity in SLE and can be used to monitor response to treatment
- Anti-lupus anticoagulant decreases the risk of nephritis in SLE

Ans. (d)



33. RHEUMATOID ARTHRITIS

R	- Test useful: RA factor - RA factor can be biologically false positive in 5% of normal population
H	- Hand joints involved: - Proximal interphalangeal - Metacarpophalangeal - Wrist joints B/L symmetrical involvement
E	Young female with early morning stiffness > 1 hour
U	Ulnar deviation of hand → indicative of degenerative joint diseases
M	Drugs used: DMARDs (Disease-Modifying Anti-Rheumatic Drugs)
A	Anti-CCP antibody (Cyclic Citrulline Peptide)
T	Typical deformities: Swan neck deformity - extension at the proximal interphalangeal joint

Important Information

- Flexion at distal interphalangeal joint, neutral position at PIP: Mallet finger
- Migratory polyarthritis:
 - Asymmetrical
 - Involves knee joint, ankle joint, elbow joint
- Early morning stiffness is seen in:
 - Ankylosing spondylitis
 - RA



Mallet finger

PATHOLOGY

00:04:44

- Synovial inflammation + proliferation
- Pannus formation**
 - Granulation tissue with proinflammatory cytokines that cause destruction of cartilage, followed by subchondral bone
- End result is **erosive arthritis**

CLINICAL FEATURES

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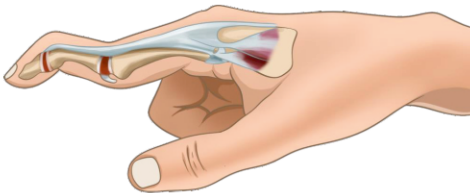
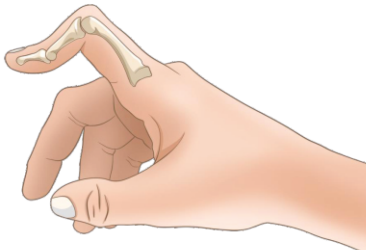
- Young female with **constitutional symptoms** (low-grade fever, myalgia, arthralgia) or an old lady with deformities
- Early morning stiffness lasting for **>1 hour** involving the small joints of hand
 - Difficulty in household chores, brushing teeth



- M/c joints involved are **bilateral PIP, MCP, wrist joints**
- **Metacarpophalangeal joints and distal interphalangeal joints** involvement
- **1st metatarsophalangeal joint** is involved in RA
 - In Gout- 1st metatarsophalangeal joints
 - H/o purine-rich diet: Alcohol binge and non-vegetarian food in large amounts for past couple of weeks
 - Patient started on thiazide diuretics (↑ level of uric acid)
 - DIP involvement → Concomitant osteoarthritis in old age
- Knee and shoulder joints
- **Flexor tendon tenosynovitis** (hallmark)
 - Hand grip ↓

DEFORMITIES

00:08:55

SWAN NECK DEFORMITY	• Flexion at distal interphalangeal joint • Extension at proximal interphalangeal joint  
BOUTONNIERE'S DEFORMITY	• Flexion at proximal interphalangeal joint • Extension at distal interphalangeal joint  
Z-LINE DEFORMITY	• Subluxation of 1st metacarpophalangeal joints • Hyperextension of interphalangeal joint 
PIANO KEY MOVEMENT OF ULNAR STYLOID	

- Risk of **atlanto axial dislocation** → pressure on spinal cord → quadriplegia
- Risk of compressive myelopathy

- Rheumatoid nodules
 - Extensor in distribution
 - **M/c extra-articular manifestation**



AUTOIMMUNE DISORDER

- Secondary Sjögren syndrome

INTERSTITIAL LUNG DISEASES

- Usual interstitial pneumonitis

- Development of **pulmonary nodules** (rheumatoid pulmonary nodules)
 - Breakdown of nodules → Exudative pleural effusion
 - **Lights criteria** for rheumatoid pleural effusion
 - Pleural fluid protein > 0.5
 - LDH > 0.6
 - Pleural Fluid LDH > 2/3 times as the standard described
- Hematological manifestations: **Anemia of Chronic diseases**
 - Normocytic normochromic anemia > Microcytic hypochromic anemia
 - Normocytic normochromic anemia presents early
 - If diagnosis is made later- microcytic hypochromic anemia
- Rheumatoid nodules: Extensor aspect of fingers, Achilles tendon, sacral prominences

ORGAN	MANIFESTATIONS
EYE	Keratoconjunctivitis sicca , scleritis (Scleromalacia perforans)
MOUTH	Xerostomia
LUNGS	• Rheumatoid nodules in lungs (inflammatory) • Evaluation of hydrothorax of patient: Exudative variety • Interstitial lung diseases- Usual interstitial Pneumonitis • Caplan syndrome: Coal workers pneumoconiosis + Rheumatoid arthritis
HEART	• Rheumatoid nodules involve outermost layer of the heart → Pericardial effusion • Ischemic heart disease: Leading cause of death • Pericarditis • Arrhythmia in case of involvement of conduction system of heart • Valvular lesion: Mitral Regurgitation (M/c)
KIDNEY	• Membranous glomerulopathy: ◦ Primary variety → Anti-phospholipid A2 receptor antibody



	○ Secondary variety → Associated with connective tissue disorders, malignancy • Development of secondary amyloidosis
	• Rheumatoid nodules on dorsum of hand, olecranon process, Achilles tendon, purpura
GENITALS AND OTHERS	• Hypoandrogenism • Osteoporosis (T score and Z score of patient < -2.5 SD)
HEMATOLOGICAL MANIFESTATIONS	• Anemia of chronic diseases: Normocytic normochromic anemia

CLASSIFICATION CRITERIA FOR RHEUMATOID ARTHRITIS

00:23:40

(Mnemonic: **JSAD**)

- **J**oint involvement: Large and small joint involvement
 - Monoarticular RA: 1 joint
 - Oligoarticular RA: <4 joints
 - Polyarticular RA: ≥5 joints
- **S**erological
 - RA factor (biologically false positive in 5% of normal population)
 - Anti-Cyclic citrullinated peptide antibody
- **A**cute phase reactants: ESR and CRP normal/↑

Important Information

Felty syndrome: RA+ Splenomegaly + Neutropenia

INVESTIGATIONS

00:26:00

- ESR, CRP → ↑
- **RA factor +: Screening test**
- **Anti-CCPA/ACPA: IOC**
- MRI/USG/ X-ray wrist: Imaging to demonstrate erosions
- Synovial fluid aspiration is not done routinely
 - In gouty arthritis: **needle-shaped crystals** in turbid fluid and under polarized microscopy → negative birefringence
- X ray foot: **5th MTP** involvement on lateral aspect

TREATMENT

00:27:01

- Non-steroidal anti-inflammatory drugs: Not first line/adjunctive role
 - Pain killers have a limited role in management, and do not prevent the progression of the erosive arthritis component
- Avoid steroids as much as possible
- DMARDS – To prevent degeneration and extra-articular symptoms
- Biological DMARDS (mnemonic: **ACE:IG**)
 - **A**dalimumab
 - **C**ertolizumab
 - **E**tanercept